1905 ▸ 1910

1905

Haber–Bosch process

German chemist Fritz Haber described a process to make a chemical called ammonia—a crucial ingredient in fertilizers—from a chemical reaction involving nitrogen and hydrogen. German chemist Carl Bosch scaled up Haber's laboratory process so that industrial quantities of ammonia could be produced.

Tractor tows a fertilizer sprayer

Learning chemistry See pages 146–147

Bakelite radio

1907

Pioneering plastic

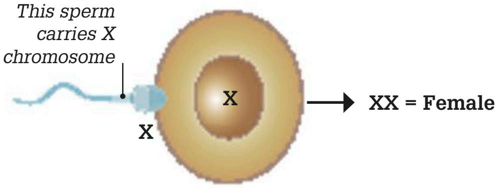
Belgian-American chemist Leo Baekeland developed a pioneering type of plastic (later named Bakelite) using a chemical called phenol, derived from coal tar and formaldehyde, derived from wood alcohol. Bakelite was cheap to produce, set hard, and had high resistance to electricity and heat. It became widely used as an electrical insulator and was molded into thousands of products—from telephones to jewelry.

1906

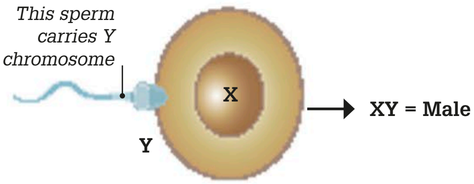
Defining allergies

Austrian physician Clemens von Pirquet defined the term "allergy." It is an overreaction in the body triggered by the body's immune system in response to something in the environment, such as dust, pollen, or certain foods it sees as being harmful.

1905



Egg carries X chromosome



Egg carries X chromosome

1905

Sex chromosomes

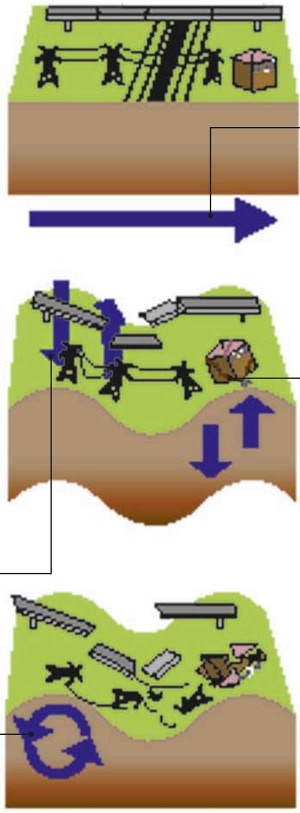
American geneticists Nettie Stevens and Edmund Beecher Wilson independently described the XX and XY system of sex chromosomes that play a role in reproduction. Sperm cells from males, which fertilize egg cells from females, either carry an X (female) or Y (male) chromosome. This joins with the X chromosome found in the egg cell to form either a male (XY) or female (XX) child.

1906

Seismic waves

British geologist Richard Dixon Oldham defined the different types of seismic wave (energetic waves caused by vibrations of moving rocks underground) that occur during earthquakes. "P," or primary, waves travel through solids, liquids, and gas at speeds of 0.8–9 miles/s (1–14 km/s); "S", or secondary, waves can only travel through solids at speeds of 0.8–5 miles/s (1–8 km/s); and surface waves, which are the slowest of all.

Types of seismic wave



P waves travel horizontally deep below the ground and are often heard but not felt. They stretch rock, sometimes causing it to fracture.

S waves may cause buildings to crack and collapse.

S waves force rock to move from side to side, at right angles to the direction of the wave's path.

Surface waves can roll and buckle the landscape, causing serious damage.

The code of life See pages 198–199